



## Extra — Nesting: Code Inside Code

**Topic:** Nesting & Indenting · **Course:** Extra Worksheet · **Time:** about 30 minutes

### 1 · Code Lives in Boxes

In block code, a loop has an arm. The arm holds the blocks inside.

Real code has no arm. So we **indent**. We push the inside lines to the right with spaces.

Indenting shows what is **inside**.

The box is the inside:

```
while true:
```

INSIDE the loop

```
    place block
```

## 2 · Inside Means "Only If" 🛡️

The hero raises a shield only **if** a monster is near.

if a monster is near:

INSIDE the if — runs **ONLY** if a monster is near

raise shield

say "pew, I am safe" ← OUTSIDE — runs **EVERY** time

Inside the box → runs only when a monster is near.

Outside the box → always runs.

**No monster is near. Which line runs? Which is skipped?**

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### 3 · Why It MUST Be Inside ⚠️

The indent tells the computer where the `if` stops.

Now we push the last line **inside** too:

`if a monster is near:`

Now EVERYTHING is inside the if

```
raise shield
```

```
say "phew, I am safe"
```

Now the hero feels safe only when a monster is near. Wrong! 🤖

Inside the box → runs only if true. Outside the box → always runs. Move a line in or out → the code changes.

Where should `say "phew, I am safe"` go? Why?

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## 4 · Predict 🧠

Read this code. The agent runs it over and over.

```
function go:
  while true:
    move forward
    if there is a flower:
      pick the flower
```

`move forward` is inside the `while`, so it happens every time.

`pick the flower` is inside the `if`, so it happens only sometimes.

**What does the agent do every time? When does it pick a flower?**

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## 5 · Debug 🐛

This code should pick a flower **only** when there is one. But the agent picks all the time. The indent is wrong.

```
function go:
  while true:
    if there is a flower:
      pick the flower
```

`pick the flower` is not pushed in under the `if`. So it is **outside** the `if`, and runs every time.

**Write the code again with the right indent. Then say why your fix works.**

## 6 · Write

Now you write one piece of code. Make the agent do this:

- a **function** called `patrol`
- inside it, a **while true**
- inside the while, an **if a monster is near**
- inside the if, **swing the sword**

Push each box in one more step than the box above it.

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**Submit** 

Send this worksheet to teacher on KakaoTalk.